

Skills Summary

Division: Finding the Unknown Factor

Use this reference while completing your worksheet or when studying.

Skill 1 — Solving a missing-factor equation

Rule: To solve $b \times n = \text{total}$, ask: “ b times *what* makes the total?”
Run through the b times table until you land on the total.

Example: Solve $6 \times n = 30$.

Step 1. Ask: 6 times what makes 30? Skip count by 6 until you reach 30.

Step 2. 6, 12, 18, 24, 30 — that is five 6s, so $6 \times 5 = 30$.

$\Rightarrow n = 5$

Try it: Solve $4 \times n = 28$.

check: $4 \times 7 = 28$

Skill 2 — Division as an unknown factor

Rule: $\text{total} \div b$ asks the same question as $b \times ? = \text{total}$.
A related multiplication fact you already know gives the answer.

Example: Find $45 \div 5$.

Step 1. Rewrite the division as a missing-factor question: $5 \times ? = 45$.

Step 2. From the fives table, $5 \times 9 = 45$, so $45 \div 5 = 9$.

$\Rightarrow 9$

Try it: Find $32 \div 4$ by thinking $4 \times ? = 32$.

check: $4 \times 8 = 32$

Skill 3 — Equal-group story problems

Rule: Find the total and the given group number first.
 $\text{total} \div \text{number in each group} = \text{how many groups}$; $\text{total} \div \text{number of groups} = \text{how many in each}$.

Example: There are 28 pencils, packed 4 to a pack. How many packs?

Step 1. The total is 28 and each group holds 4, so ask: 4 times what makes 28?

Step 2. $4 \times 7 = 28$, so $28 \div 4 = 7$.

$\Rightarrow 7$ packs

Try it: There are 54 books shared equally on 6 shelves. How many books on each shelf?

check: $6 \times 9 = 54$

(!) Watch out: Read carefully *which* number the story gives you — the number of groups or the number in each group. The division is the same, but the unit of your answer (bags? apples?) comes from what the question asks.